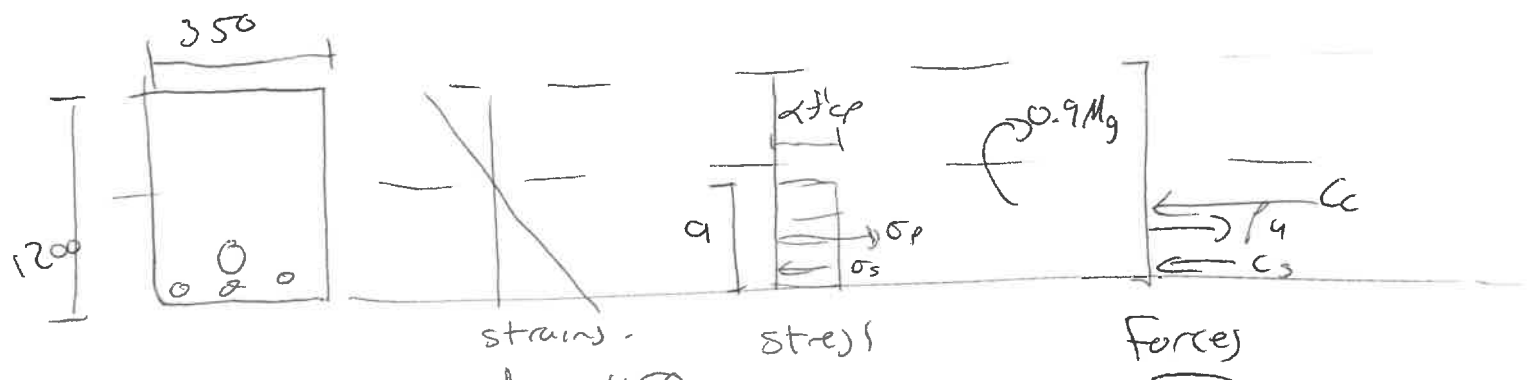


Lecture Example

Strength at transfer



$$A_{st} = 3000 \text{ mm}^2 \quad \cancel{d_s = 1150}$$

$$f_{sy} = 500 \text{ MPa}$$

$$0.9 \times M_g = 738 \text{ kNm}$$

$$f'_{cp} = 32 \text{ MPa}, \alpha = 0.8$$

$$\cancel{d_p = 1150}$$

$$C_s = 3000 \times 500 = 1,500,000 \text{ N}$$

$$C_c = \alpha f_{cp} \times a \times b = 0.8 \times 32 \times a \times 350 = 8960a$$

$$\Sigma F = 0 \quad P_u = C_c + C_s$$

$$P_u = 1.5 \times 10^6 + 8960a \quad \text{--- (1)}$$

Moment equilibrium @ bottom fibre of beam

$$C_s \times d_s + C_c \times \frac{a}{2} = P_u \cdot d_p = 0.9 M_g$$

$$C_s \times 50 + C_c \frac{a}{2} = 150 P_u + 0.9 M_g$$

$$\left[\begin{array}{l} d_p = 150 \text{ mm} \\ d_s = 50 \text{ mm} \end{array} \right.$$

$$75 \times 10^6 + \frac{8960 a^2}{2} = 225 \times 10^6 + 1.34 \times 10^6 a + 738 \times 10^6$$

$$4480 a^2 - 1.34 \times 10^6 a - 8.9 \times 10^8 = 0 \quad \text{--- (2)}$$

solving quadratic

$$\Rightarrow a = 619 \text{ mm}$$

$$P_u = 7046 \text{ kN}$$

$$0.6 P_u > 1.15 P_j$$

$$0.6 \times 7046 \geq 1.15 P_j$$

$$\Rightarrow \underline{P_j < 3676 \text{ kN}}$$

EXAMPLE 7.1

CALCULATE THE LOAD ON A POST-TENSIONED BEAM IN SHEAR FOR A WEB-SHEAR CRACK TO FORM

A post-tensioned beam with the cross-section shown in Figure 7.8 is to span 25 metres between simple supports and carry a uniform live load of 24 kN/m. The beam is prestressed at each end by a parabolic cable with eccentricity of 524 mm at mid-span and zero at the ends. The cable comprises 22 strands of 12.7 mm diameter with $A_p = 2170 \text{ mm}^2$, in a 112 mm diameter duct. Owing to friction the prestressing force varies in an approximately linear manner from 2435 kN at each end to 2300 kN at mid-span.

The beam also has longitudinal reinforcement consisting of 5-N28 bars ($A_{st} = 3100 \text{ mm}^2$) at depth $d_{st} = 1160 \text{ mm}$, throughout the span. For the longitudinal reinforcing steel and the stirrups we take $f_{sy} = 500 \text{ MPa}$ and $f_{sy,f} = 250 \text{ MPa}$ (Grade R). The concrete strength is $f'_c = 40 \text{ MPa}$.

Calculate the shear corresponding to the formation of a significant web-shear crack at a distance of $x = 1.0$ metres from the support.

Other data:

$$A_g = 488.4 \cdot 10^3 \text{ mm}^2; I_g = 83.8 \cdot 10^9 \text{ mm}^4$$

$$y_t = 552 \text{ mm}; y_b = 668 \text{ mm}$$

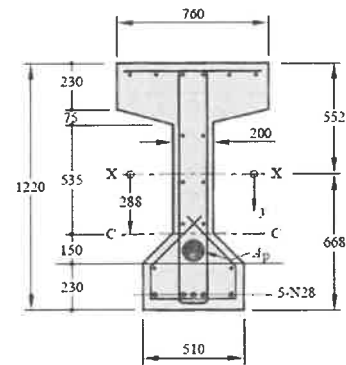
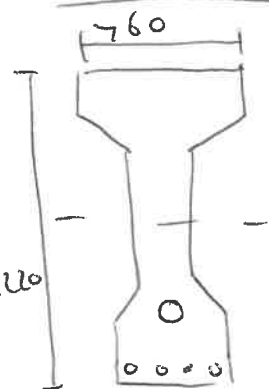


Figure 7.8 Cross-section for Examples 7.1, 7.2 and 7.3

Lecture Problem.

Ex 7.1 Calculate load on PCB in shear for web shear crack to form.



$$e \text{ @ mid span} = 524 \text{ mm}$$

$$A_p = 2170 \text{ mm}^2$$

$$A_{st} = 5N28 = 3100 \text{ mm}^2, \text{ dist} = 1160 \text{ mm}$$

$$\text{Span} = 25 \text{ m}$$

$$A_g = 488.4 \times 10^3 \text{ mm}^2 \quad I_g = 83.8 \times 10^9$$

$$y_c = 552 \quad y_b = 668 \text{ mm}$$

@ $x = 1 \text{ metre}$ from support.

$$\text{Self-weight} = A_g \times \gamma_c = 488.4 \times 10^3 \times 25 \times 10^{-6} = 12.2 \text{ kN/m}$$

$$\text{service load} = W_s = 12.2 + 24 = 36.2 \text{ kN/m}$$

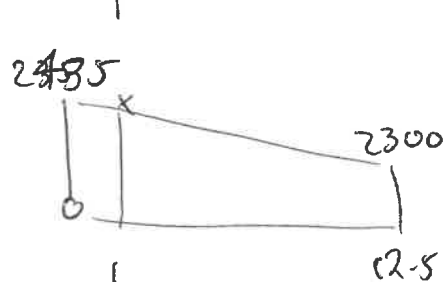
$$\begin{array}{c} \text{36.2 kN/m} \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ \uparrow \quad \quad \quad x \\ R_L = 36.2 \times \frac{25}{2} \\ = 452.5 \text{ kN} \end{array}$$

$$V^* = 452.5 - 36.2x \text{ (kN)}$$

$$M^* = 452.5x - 18.1x^2 \text{ (kNm)}$$

$$\Rightarrow x = 1 \text{ m, cable force} = \underline{2424 \text{ kN}}$$

i.e. prestress force variation due to friction loss



$$\frac{2435 - 2300}{12.5} = \frac{2435 - x}{1.0}$$

$$V(x=1) = 416 \text{ kN}$$

$$M(x=1) = 438 \text{ kNm}$$

Sign-convention: for Mohr's circle calculation
treat tension as positive

Web-shear cracking

shear carried by concrete at web-shear cracking, V_E , is that shear for which the principle stress at some point in the web reaches critical value
 at $f_{ct} = 0.36 \sqrt{f_k} = 0.36 \sqrt{40} \text{ MPa} = 2.28 \text{ MPa}$
 (tensile)

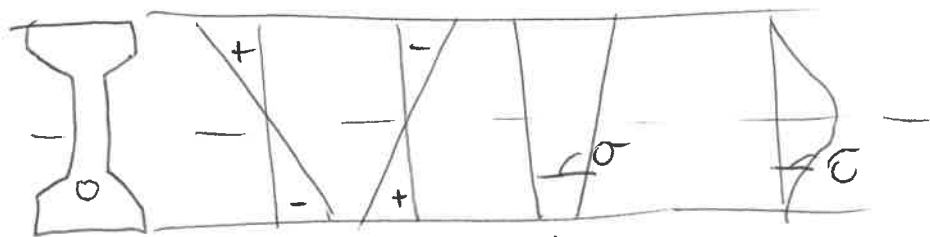
- at centroid ($y=0$), the horizontal normal stress

$$\sigma_x = -\frac{P}{A_g} = \frac{-2424 \times 10^3}{488.4 \times 10^3} = -4.96 \text{ MPa} \quad (\text{comp})$$

- the vertical normal stress is assumed to be zero

- the shear stress

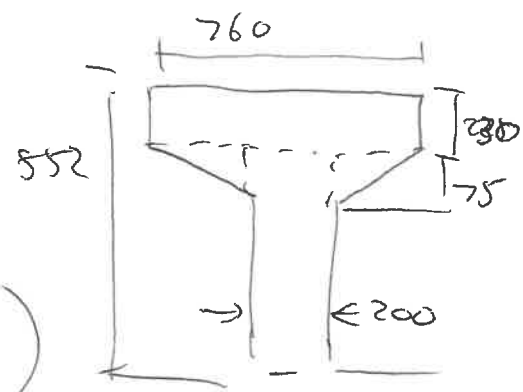
$$\tau = \frac{V_E Q}{I b_w}$$



$$\sigma = -\frac{P}{A} - \frac{P \cdot e \cdot y}{I} + \frac{M_y}{I} \quad \begin{matrix} \text{resultant} \\ \text{stress} \end{matrix} \quad \begin{matrix} \text{shear stress} \end{matrix}$$

Q is the first moment of area above (or below) the centroid.

$$\begin{aligned}
 Q &= 760 \times 230 \left(552 - \frac{230}{2} \right) \\
 &\quad + \\
 &\quad 75 \times \left(\frac{760 - 200}{2} \right) \times \left(552 - 230 - \frac{75}{3} \right) \\
 &\quad + \\
 &\quad 200 \times (552 - 230) \times \left(\frac{552 - 230}{2} \right) \\
 &= 92.99 \times 10^6 \text{ mm}^3
 \end{aligned}$$



⇒ The shear stress in the web is then

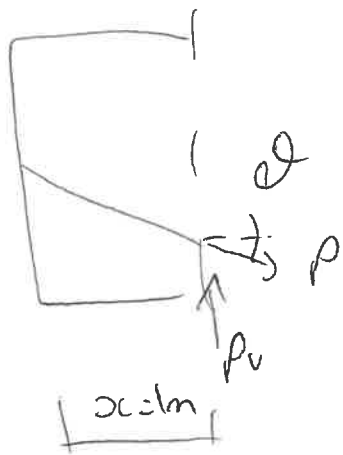
$$\tau_{xy} = \frac{V_E \times 92.99 \times 10^6}{83.8 \times 10^4 \times 144} = 7.71 \times 10^{-6} V_E \text{ (MPa)}$$

taking $\sigma_i = \text{allow } f_{ct} = 2.28 \text{ MPa}$.

$$\begin{aligned}
 \sigma_1 &= \sqrt{(0.5 \sigma_c)^2 + \tau_{xy}^2} + 0.5 \sigma_x \\
 &= \sqrt{0.5(-4.96)^2 + (7.71 \times 10^{-6} V_E)^2} + 0.5(-4.96) \\
 &= 2.28 \text{ MPa (tensile)}
 \end{aligned}$$

solving gives $V_E = 527 \times 10^3 \text{ N}$ (527 kN)

⇒ Vertical component of the cable force
 $P_v = P \sin \theta = P_v \cdot \theta$



→ eccentricity of cable @ $x=1m$

$$y = 4h \left(\frac{x}{L} + \left(\frac{x}{L} \right)^2 \right)$$

$$y = 4 \times 524 \left(\frac{1000}{25000} + \left(\frac{1000}{25000} \right)^2 \right)$$

$$y = 80.5mm = e$$

→ the inclination of the cable is

$$\theta = \frac{4h}{L} \left(1 - 2 \frac{x}{L} \right) = \frac{4 \times 524}{25000} \left(1 - 2 \frac{1000}{25000} \right)$$

$$= 0.077$$

$$\therefore P_v = 2424 \times 0.077 = 187kN$$

$$V_{cr} = V_t + P_v = 527 + 187 = 714kN > V^k 416$$

∴ ok for web crushing.

EXAMPLE 7.3**DESIGN OF POST-TENSIONED BEAM FOR SHEAR**

Design the shear reinforcement for the beam given in Example 7.1. For the stirrups use Grade R for which $f_{sv,t} = 250$ MPa. The tendon consists of 22 strands of 12.7 mm diameter and from Table 2.1 $f_{pb} = 1870$ MPa and $E_p = 195$ GPa. The cross-section is shown in Figure 7.8.

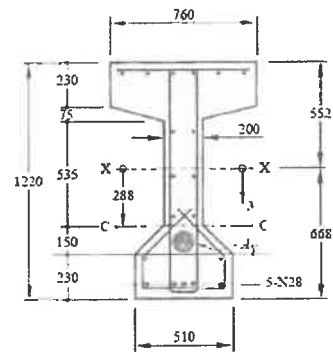


Figure 7.8 Cross-section for Examples 7.1, 7.2 and 7.3

Lecture Problem

7.3. Design for shear

⇒ Design load

$$w^* = 1.2 \times 12.2 + 1.5 \times 24 = 50.6 \text{ kN/m}$$

$$V^*(x) = 633 - 50.6x \text{ (kN)}$$

$$M^*(x) = 633x - 25.3x^2 \text{ (kNm)}$$

⇒ force in cable and slope @ distance x from support.

$$P_c = 2435 - 10.8x \text{ (kN)}$$

$$\text{slope } \alpha(x) = 4 \times \frac{524 \times 10^{-3}}{25} \left(1 - \frac{2x}{25} \right)$$

$$= 0.0838 \left(1 - \frac{x}{12.5} \right) \text{ radians}$$

⇒ Depth of tendon at distance x from support

$$e(x) = \frac{524 \times 10^{-3}}{11.93} \left[\frac{x}{25} - \left(\frac{x}{25} \right)^2 \right]$$

$$= \frac{x}{11.93} - \frac{x^2}{248.2} \text{ (m)}$$

⇒ shear design at a section at distance d_v from support

$$d_v = 0.72 \times D = 0.7 \times 1220 = \underline{878 \text{ mm}}$$

$$\text{take } \alpha = 0.878 \text{ m}$$

⇒ effective web width

$$b_v = 200 - (6.5 \times 112) = 147 \text{ mm}$$

⇒ vertical component of cable force

$$P_v = P \times \alpha = 2426 \times 0.0779 \\ = 189 \text{ kN}$$

$$V^* = 633 - 50.6 \times 0.878 = 589 \text{ kN}$$

$$M^* = 633 \times 0.878 - 25.3 \times 0.878^2 \\ = 536 \text{ kNm}$$

⇒ Calculate concrete contribution to shear capacity

Cl. 8.2.4.1

$$\phi_p = 0.9$$

$$K_v = \left[\frac{0.4}{1 + 1500 \epsilon \alpha} \right]$$

$$\epsilon \alpha = \left[\frac{M^*}{dv} + V^* \right] \neq \phi_p P_v + 0.5 N^* - A_{pt} f_{po}$$

$$2 (E_s A_{st} + E_p A_{pt})$$

$$f_{po} = 0.5 f_{pb}$$

within $0 \leq \epsilon \alpha \leq 3 \times 10^{-3}$

$$\epsilon \alpha = \left[\frac{536 \times 10^6}{878} + 589 \times 10^3 \right] - 0.9 \times 189 \times 10^3 - \frac{2170 \times 0.5 \times 1870}{2}$$

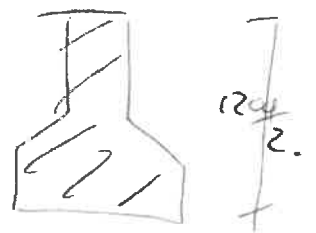
$$2 (200 \times 10^3 \times 3100 + 795 \times 10^3 \times 2170)$$

$$= -479 \times 10^{-6}$$

as calculated value of ϵ_x is less than zero, we shall recalculate using

$$\epsilon_x = \frac{\left[\frac{M^*}{dv} + V^* \right] + \gamma_p P_v + 0.5 N^* - A_{pt} f_{pb}}{2 \left[E_s A_{st} + E_p A_{pt} + E_c A_{ct} \right]}$$

$$A_{ct} = \frac{1200}{2} \times 200 + 230 (510 - 200) + \frac{150}{2} (510 - 200) \\ = 216.6 \times 10^3 \text{ mm}^2$$



$$\epsilon_{xc} = \frac{-999.6 \times 10^3}{2 \times (1043 \times 10^9 + (32.8 \times 10^3 \times 216.6 \times 10^3))} \\ = -61.3 \times 10^{-6} \text{ and greater than } -0.2 \times 10^{-3}$$

$$\therefore K_v = \frac{0.4}{1 + 1500 \times (-61.3 \times 10^{-6})} = 0.441$$

$$\phi_v = 29 + 7000 \times (-61.3 \times 10^{-6}) = 28.6^\circ$$

$$\Rightarrow V_{uc} = K_v b_v dv \sqrt{f_k} \\ = 0.441 \times 144 \times 878 \times \sqrt{40} \\ = 352.6 \text{ kN}$$

$$\phi V_{uc} = 0.75 \times 352.6 = 264 \text{ kN}$$

⇒ Design for shear reinforcement
the minimum shear that is required to take
by stirrups

$$V_{us} \geq \frac{1}{4} [V^* - (\phi W_{uc} + \delta_p P_v)]$$

$$= \frac{1}{0.75} [589 - (264 + 0.9 \times 189)]$$

$$= 207 \text{ kN}$$

⇒ the area of shear reinforcement
per spacing is

$$A_{sv}/s = V_{us} / (f_{syf} \cdot d_v \cdot \cot \alpha_v) - \text{Cl 8.2.5.2}$$

$$A_{sv}/s = 207 \times 10^3 / (250 \times 878 \times \cot 28.6)$$

$$= 0.514$$

check minimum web reinforcement

$$D > 750 \therefore \frac{A_{sv, \min}}{s} = 0.08 \sqrt{40} \times 144 / 250$$

$$= 0.291 < 0.514$$

$$\text{Hence } A_{sv}/s \geq 0.514, A_{sv} = 2 \times 80 = 160 \text{ mm}^2 \quad (\text{R10})$$

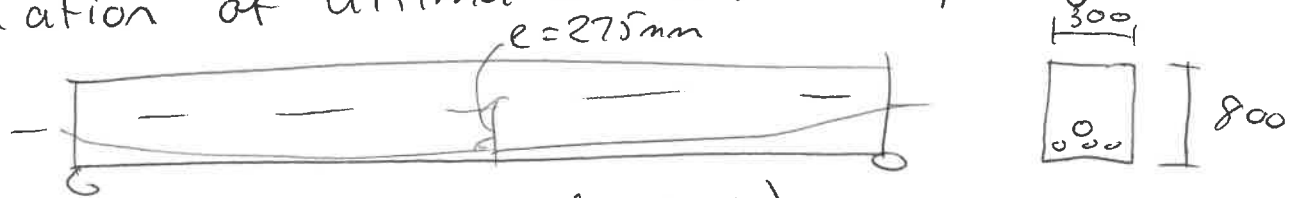
$$s \leq 160 / 0.514 = 311 \text{ mm}$$

adopt R10 @ 300 spacing - max spacing = 300

③

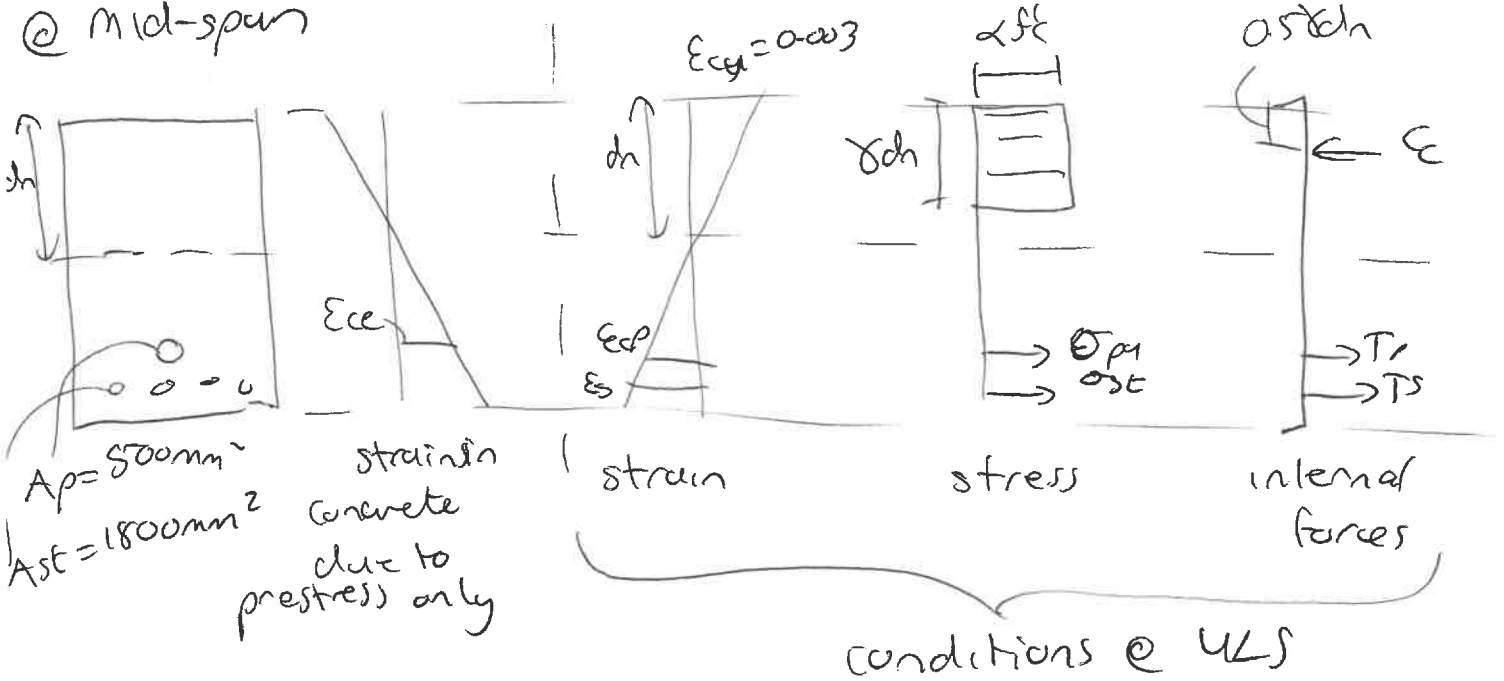
Lecture Example

⇒ Calculation of ultimate moment capacity M_u



$$f_{lc} = 32 \text{ MPa} \quad \alpha = 0.8 \quad \delta = 0.89 \quad (\text{Cl 8.1.3})$$

@ Mid-span



⇒ determine, d_n

$$\text{force equilibrium} = C_c = T_p + T_s$$

assume all steels have yielded

$$\sigma_{pu} = f_{py} = 1600 \text{ MPa} \quad \sigma_{su} = f_{sy} = 400 \text{ MPa}$$

$$T_s = A_{st} \cdot f_{sy} = 1800 \times 400 = 720,000 \text{ N}$$

$$T_p = A_p \cdot f_{py} = 500 \times 1600 = 800,000 \text{ N}$$

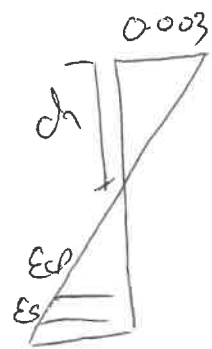
$$C_c = T_s + T_p$$

$$\alpha f_c \times \delta d_n \times b = 1520,000$$

$$\delta d_n = 1520,000 / (0.8 \times 32 \times 300)$$

$$\delta d_n = 198 \text{ mm} - (\text{depth of effective stress block})$$

⇒ check assumptions steels have yielded



$$d_n = \frac{198}{0.89} = 222 \text{ mm}$$

* steel strain, $\epsilon_{st} = \frac{E_{cu} (d_{st} - d_n)}{d_n}$

$$= 0.007 > 0.002$$

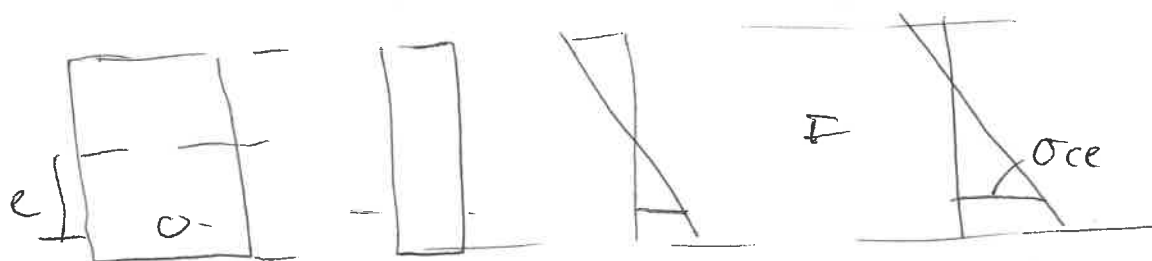
∴ tensile steel has yielded

$$\left(\frac{f_{sy}}{E_s} = \frac{400}{2 \times 10^5} \right)$$

* check strain in prestressing steel

total strain ϵ level = $\epsilon_p = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$
of prestress

ϵ_{ce} - due to prestress only



$$\sigma_{ce} = \frac{P}{A_g} + \frac{P \cdot e \cdot e}{I_g}$$

$$P = \sigma_{pe} \times A_p$$

$$= 1000 \times 500$$

$$= 500,000 \text{ N}$$

$$\sigma_{ce} = \frac{500 \times 10^3}{(800 \times 300)} + \frac{500 \times 10^3 \times (275)^2}{12.8 \times 10^9}$$

$$= 2.08 + 2.95 = 5.04 \text{ MPa}$$

$$\epsilon_{ce} = \frac{\sigma_{ce}}{E_c} = \frac{5.04}{28,570} = 0.000176$$

ϵ_{pe} - strain in prestress steel

$$= \frac{\sigma_{pe}}{E_p} = \frac{1000}{190 \times 10^3} = \underline{0.0053}$$

ϵ_{cp} - strain in concrete @ level of prestress due to applied M.

$$= \frac{\epsilon_{cu} (d_p - d_n)}{d_n} = \frac{0.003 (625 - 222)}{222}$$
$$= \underline{0.0061}$$

* Total strain @ level of prestress

$$\epsilon_{pu} = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$$

$$= 0.00176 + 0.0053 + 0.0061$$

$$= 0.0116 > 0.0084 \quad (\epsilon_{py} = \frac{f_{py}}{E_p})$$

$\therefore$ prestress steel has yielded $= \frac{1600}{190,000}$

$\Rightarrow$ Calculate Moment Capacity, M_u

$$M_u = T_s d_s + T_p d_p - C_c d_c - \#$$

$$= (720,000 \times 750) + (800,000 \times 625)$$

$$- \left[(0.89 \times 222 \times 300 \times 0.8 \times 32) \times \left(\frac{0.8 \times 0.89}{222} \right) \right]$$

$$= 930 \times 10^6 \text{ Nmm}$$

$$= 930 \text{ kNm}$$

$$\phi M_u = 0.8 \times 930$$

$$= \underline{744 \text{ kNm}}$$